Date of announcement: 28/10/22

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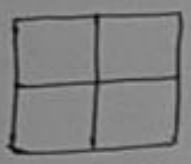
ASSIGNMENT-2

6/6

5. Explain the N-Queen's problem and its solution using Backtracking.(6 Marks)

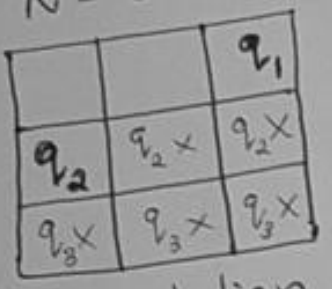
→N-Queens problem is to place n-queens in such a manner on an nxn chessboard that no queens attack each other by being in the same row, column or diagonal. for n=1, the problem has a trivial solution, and no solution exists for n=2 & n=3

N = 2.



No solution

N = 3



No solution

→So first we will consider the 4 queens problem and then generate it to n-queens problem. Given a 4x4 chessboard and numbers of rows & column of the chessboard 1 through 4.

→Since, we have to place 4 queens such as q_1, q_2, q_3 & q_4 on the chessboard, such that no two queens attack each other. Each queen must be placed on a different row. ie; we put queen "i" on row "i".

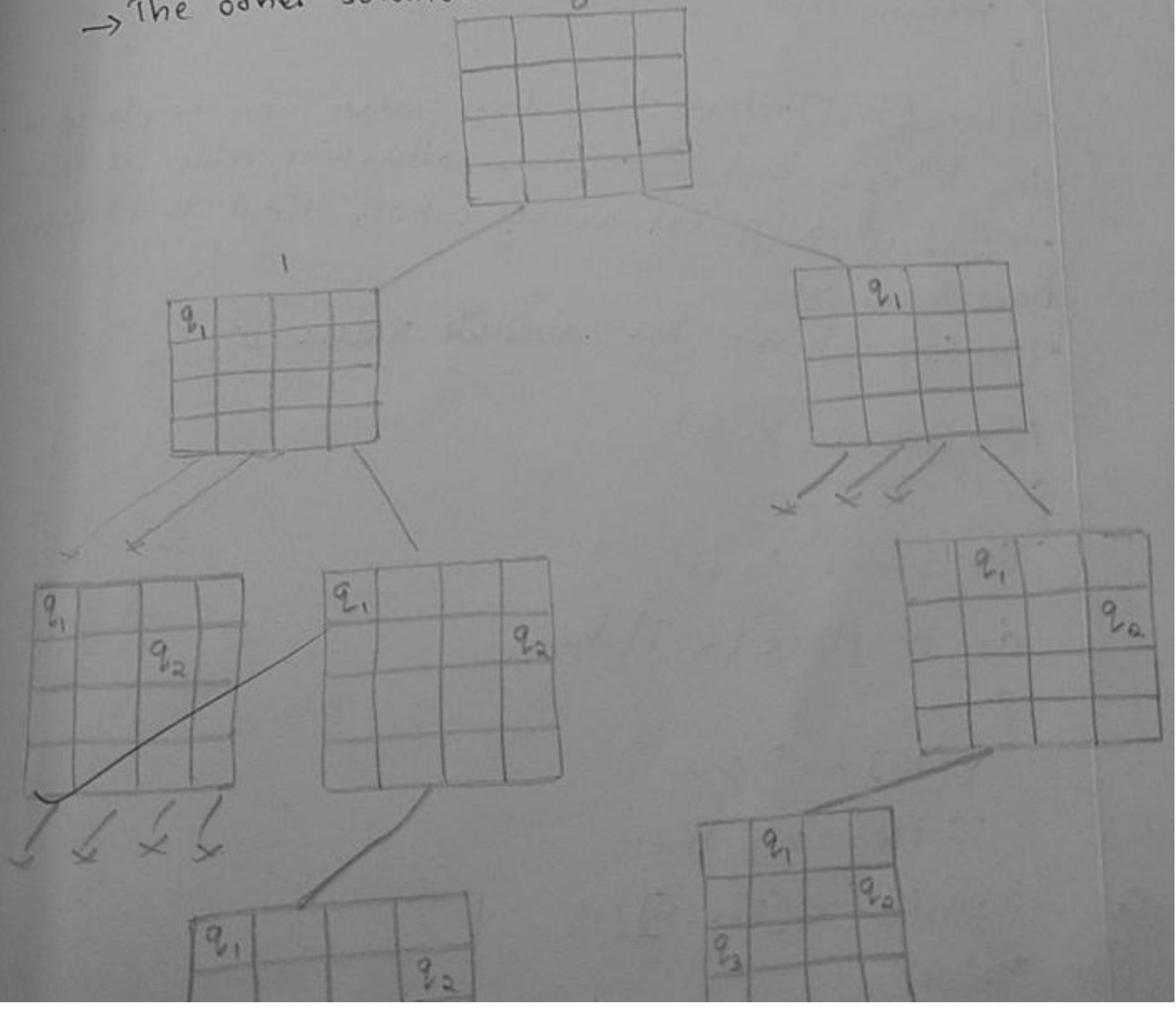
Solution

→We place queen q_1 in the very first acceptable position (1,1).

→Next, we put queen q_2 so that both these queens do not attack each other.

→We find that if we place q_2 in column 1 & 2, then this

→ in (2, 4) the next best possible solution.
→ Then we obtain the position for placing 'q_3' which is (3, 2)
→ But later this position also leads to a dead end, and no place is found where 'q_4' can be placed safely.
→ Then we have to backtrack till 'q_2' and place it to (1, 2) and then all other queens are placed safely by moving q_2 to (2, 4) q_3 to (3, 1) & q_4 to (4, 3)
→ That is we get the solution (2, 4, 1, 3). This is one possible solution for the 4 queen problem.
→ for another possible solution the whole method is repeated for all partial solutions.
→ The other solutions for 4 queens problem is (3, 1, 4, 2)



Algorithm

- Place (k, i) returns true if the kth queen can be placed in column i if it tests both whether i is distinct from all previous costs $x_1, x_2 \ldots x_{k-1}$ and whether there is no other queen on the same diagonal.
- Using place, we give a precise solution to the n-queens problems.

```
place (k, i)
{
for j ← 1 to k-1
do if (x[j] = i)
or (Abs x([j] - i)) = (Abs (j - k))
then return false;
return true;
}
```

- Place (k, i) return true if a queen can be placed in the kth row and i^{th} column otherwise return is false.
- X[ ] is a global array whose final k-1 values have been set
- Abs (r) returns the absolute value of r.

```
N-Queen (k, n)
{
for i ← 1 to n
do if place (k, i) then
{
x[k] ← i;
if (k==n) then
write (x[1...n]);
else
N-Queens (k+1, n);
}
}
```